

PROGRAM 24:

```
vedsk0831@DESKTOP-83FU7HB:~$ cd ~
cd OSSP
mkdir -p Program23
cd Program23
vedsk0831@DESKTOP-83FU7HB:~/OSSP/Program23$ pwd
/home/vedsk0831/OSSP/Program23
vedsk0831@DESKTOP-83FU7HB:~/OSSP/Program23$ nano program23.c
vedsk0831@DESKTOP-83FU7HB:~/OSSP/Program23$ gcc -Wall -Wextra -O2 program23.c -o program23
vedsk0831@DESKTOP-83FU7HB:~/OSSP/Program23$ ls -l
total 20
-rwxr-xr-x 1 vedsk0831 vedsk0831 16344 Oct  6 14:09 program23
-rw-r--r-- 1 vedsk0831 vedsk0831  1876 Oct  6 14:09 program23.c
vedsk0831@DESKTOP-83FU7HB:~/OSSP/Program23$ ./program23
```

```
=====
OSSP PROGRAM 23
Large Pipeline & Performance Test
=====
```

Launching 4 jobs...

```
Job 1 started | PID = 558
Job 2 started | PID = 559
Job 3 started | PID = 560
```

Parent waiting for all jobs...

```
Job 4 started | PID = 561
Job 1 completed | PID = 558
Job 1 finished successfully with status 0
Job 2 completed | PID = 559
Job 2 finished successfully with status 0
Job 3 completed | PID = 560
Job 3 finished successfully with status 0
```

```
Job 4 completed | PID = 575
Job 4 finished successfully with status 0
```

```
=====
All jobs have been processed.
Performance test completed.
=====
```

```
real    0m4.005s
user    0m0.004s
sys     0m0.004s
```

```
vedsk0831@DESKTOP-83FU7HB:~/OSSP/Program23$ top
top - 14:18:20 up 1:21, 1 user, load average: 0.00, 0.04, 0.05
Tasks: 25 total, 1 running, 24 sleeping, 0 stopped, 0 zombie
%Cpu(s):  0.0/0.0   0[
MiB Mem : 3801.6 total, 3120.1 free, 522.9 used, 294.0 buff/cache
MiB Swap: 1024.0 total, 1024.0 free,  0.0 used. 3278.8 avail Mem
```

PID	USER	PR	NI	VIRT	RES	SHR	S	%CPU	%MEM	TIME+	COMMAND
-----	------	----	----	------	-----	-----	---	------	------	-------	---------

```
vedsk0831@DESKTOP-83FU7HB:~/OSSP/Program23$ cd ~/OSSP/Program23
vedsk0831@DESKTOP-83FU7HB:~/OSSP/Program23$ ./program23
=====
OSSP PROGRAM 23
Large Pipeline & Performance Test
=====

Launching 4 jobs...

Job 1 started | PID = 670
Job 2 started | PID = 671
Job 3 started | PID = 672

Parent waiting for all jobs...

Job 4 started | PID = 673
Job 1 completed | PID = 670
Job 1 finished successfully with status 0
Job 2 completed | PID = 671
Job 2 finished successfully with status 0
Job 3 completed | PID = 672
Job 3 finished successfully with status 0
Job 4 completed | PID = 673
Job 4 finished successfully with status 0

=====
All jobs have been processed.
Performance test completed.
=====
vedsk0831@DESKTOP-83FU7HB:~/OSSP/Program23$ time ./program23
=====
OSSP PROGRAM 23
Large Pipeline & Performance Test
```

```

=====
OSSP PROGRAM 23
Large Pipeline & Performance Test
=====

Launching 4 jobs...

Job 1 started | PID = 687
Job 2 started | PID = 688
Job 3 started | PID = 689

Parent waiting for all jobs...

Job 4 started | PID = 690
Job 1 completed | PID = 687
Job 1 finished successfully with status 0
Job 2 completed | PID = 688
Job 2 finished successfully with status 0
Job 3 completed | PID = 689
Job 3 finished successfully with status 0
Job 4 completed | PID = 690
Job 4 finished successfully with status 0

=====
All jobs have been processed.
Performance test completed.
=====

real    0m4.006s
user    0m0.002s
sys     0m0.007s
vedsk0831@DESKTOP-83FU7HB:~/OSSP/Program23$ |

```

CODE:

```

#include <stdio.h>

#include <stdlib.h>

#include <unistd.h>

#include <sys/wait.h>

#include <sys/types.h>

#include <time.h>

```

```
#define JOBS 4
```

```
int main()
```

```
{
```

```
    pid_t pids[JOBS];
```

```
    printf("=====\n");
```

```
    printf(" OSSP PROGRAM 23\n");
```

```
    printf(" Large Pipeline & Performance Test\n");
```

```
    printf("=====\n\n");
```

```
    printf("Launching %d jobs...\n\n", JOBS);
```

```
    /*
```

```
     * Launch multiple jobs
```

```
    */
```

```
    for (int i = 0; i < JOBS; i++) {
```

```
        pids[i] = fork();
```

```
        if (pids[i] < 0) {
```

```
            perror("fork");
```

```
            return 1;
```

```
        }
```

```
    /*
```

```

    * Child process

    */
    if (pids[i] == 0) {

        printf("Job %d started | PID = %d\n",
            i + 1, getpid());

        /*

        * Simulate different workloads

        */
        sleep(i + 1);

        printf("Job %d completed | PID = %d\n",
            i + 1, getpid());

        exit(0);
    }
}

/*

* Parent waits for all jobs

*/
printf("\nParent waiting for all jobs...\n\n");

for (int i = 0; i < JOBS; i++) {

    int status;

```

```

if (waitpid(pids[i], &status, 0) < 0) {

    perror("waitpid");
    continue;
}

if (WIFEXITED(status)) {

    printf("Job %d finished successfully "
           "with status %d\n",
           i + 1,
           WEXITSTATUS(status));

} else if (WIFSIGNALED(status)) {

    printf("Job %d failed with signal %d\n",
           i + 1,
           WTERMSIG(status));
}
}

printf("\n=====\\n");
printf("All jobs have been processed.\\n");
printf("Performance test completed.\\n");
printf("=====\\n");

return 0;
}

```

